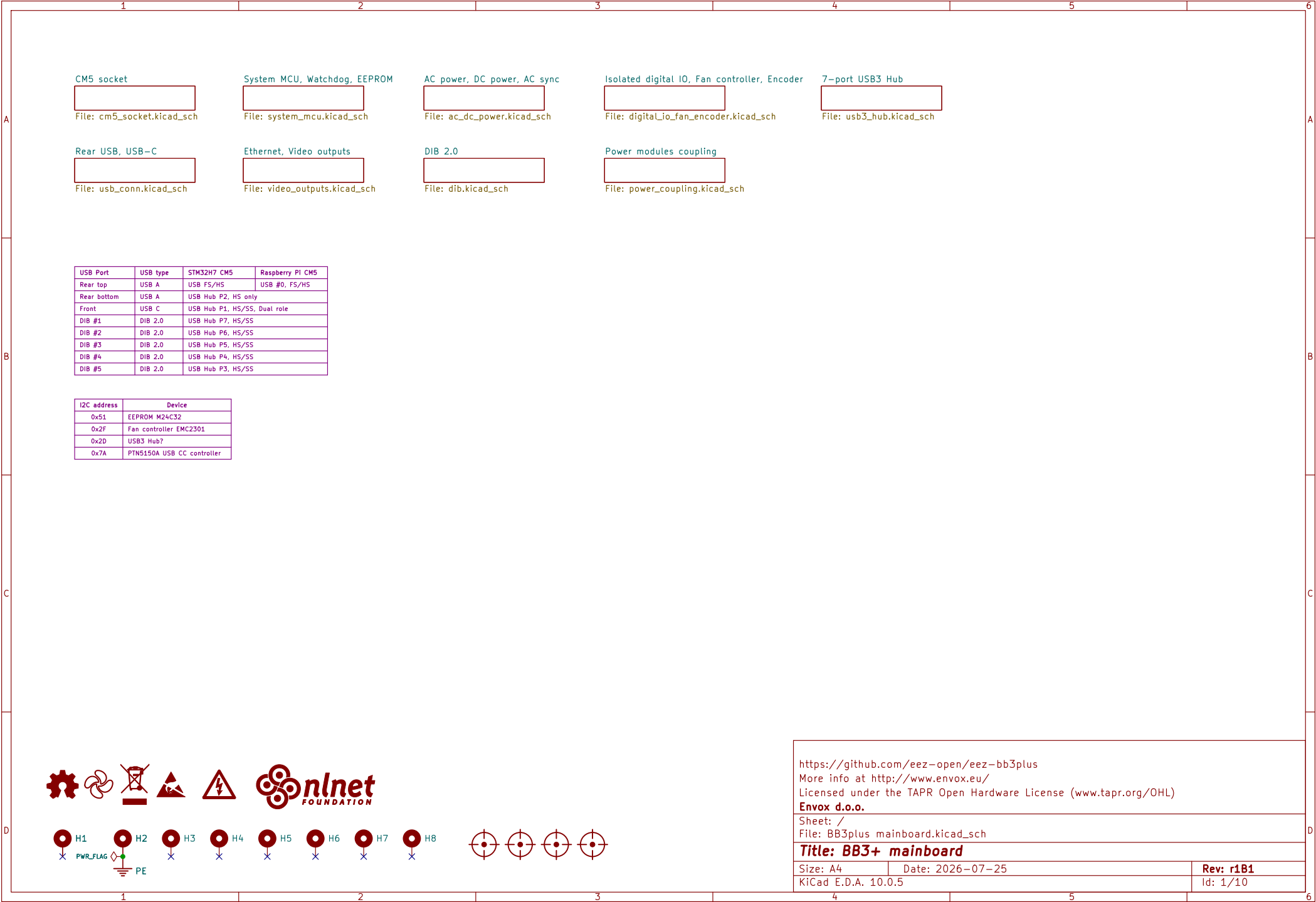




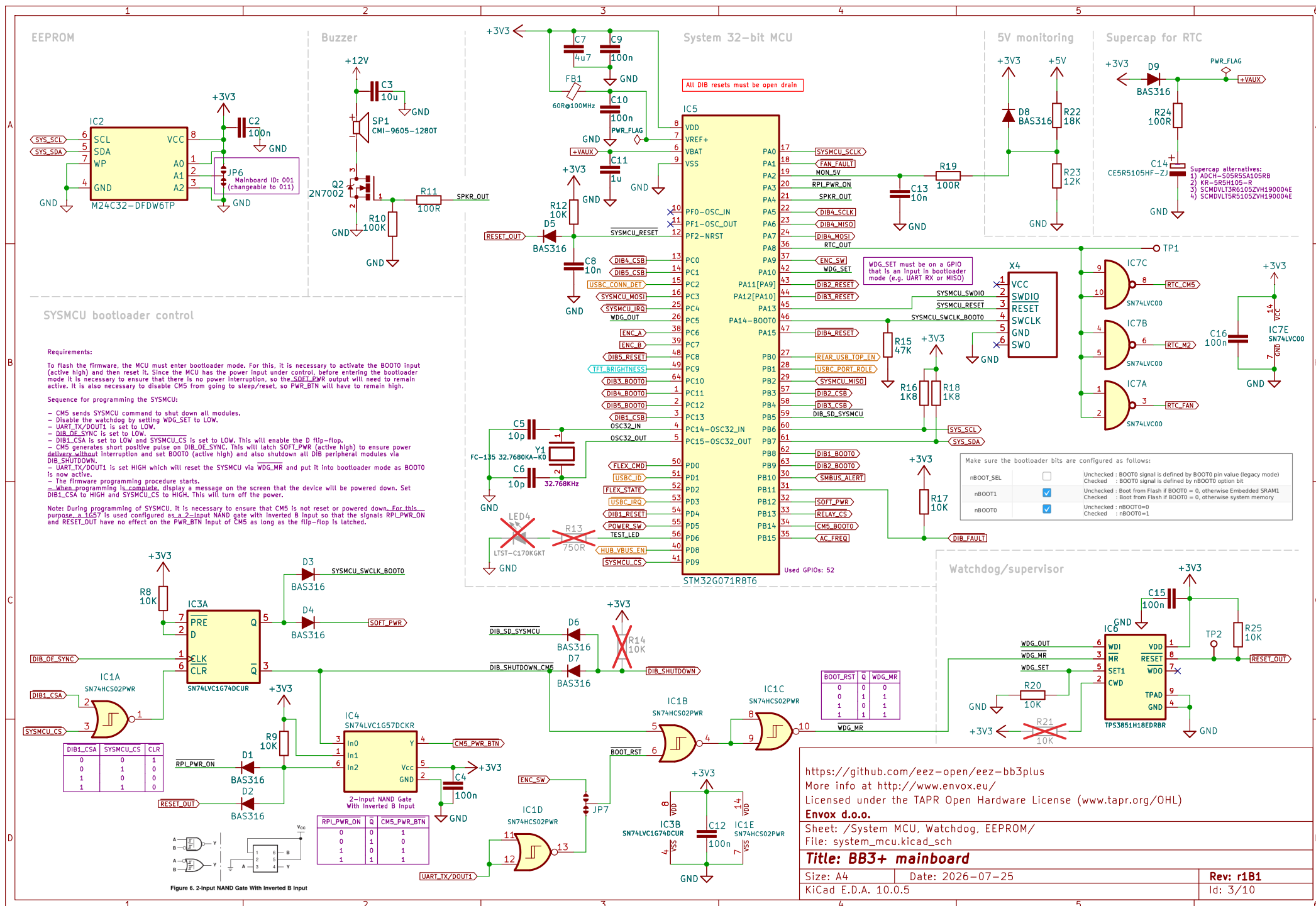
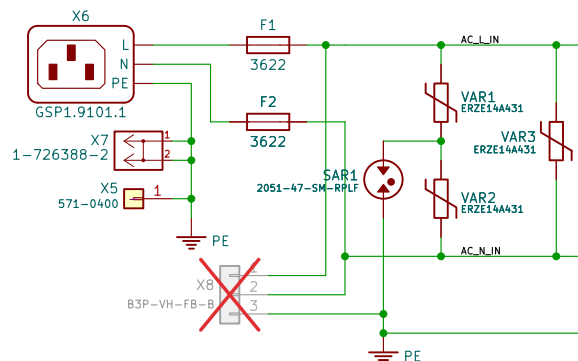
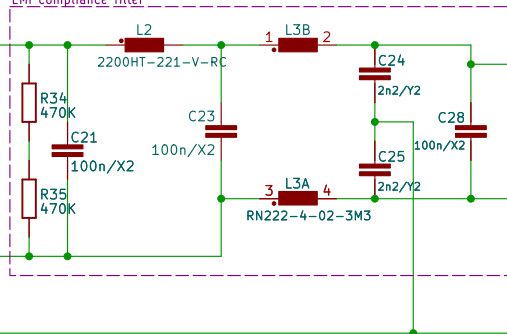


Figure 6. 2-Input NAND Gate With Inverted B Input

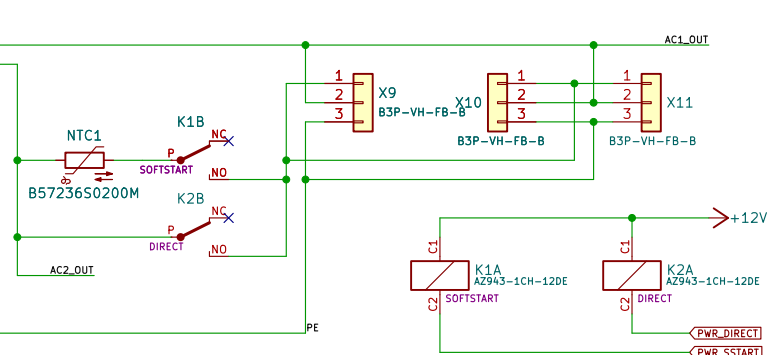
AC inlet, filtering and protection



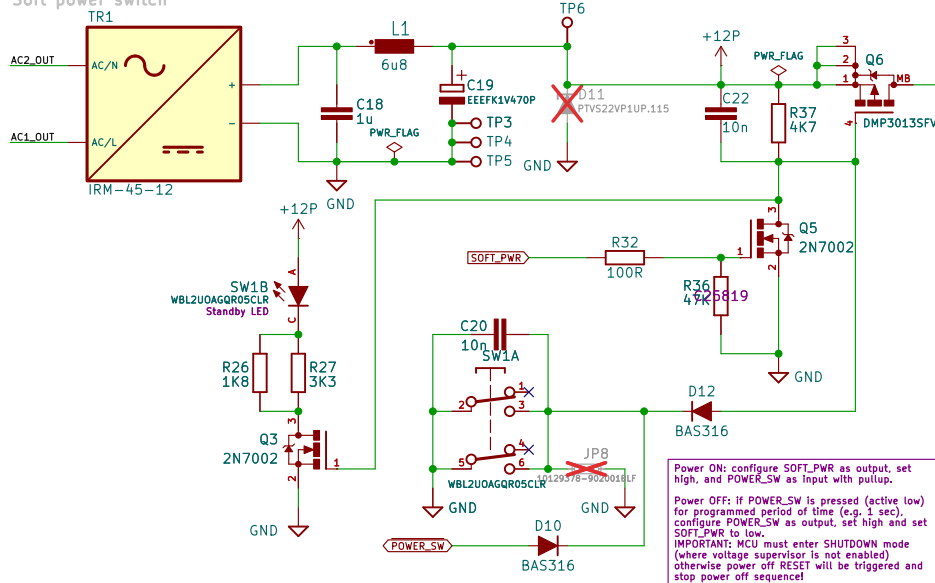
EMI compliance filter



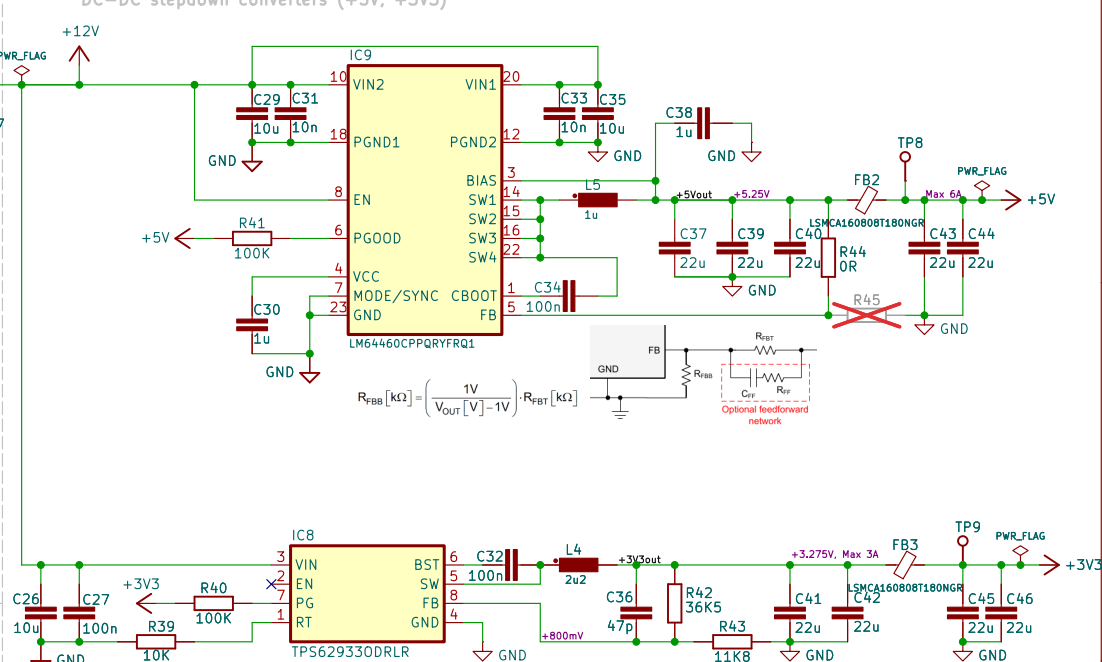
AC power control, Inrush current limiter



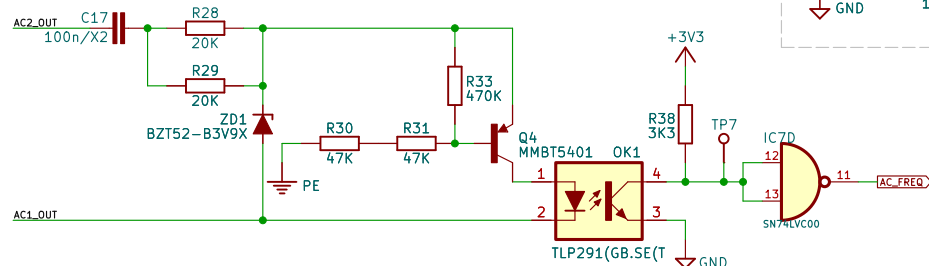
Soft power switch



DC-DC stepdown converters (+5V, +3V3)



AC Mains frequency and PE fault detection



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /AC power, DC power, AC sync/

File: ac_dc_power.kicad_sch

Title: BB3+ mainboard

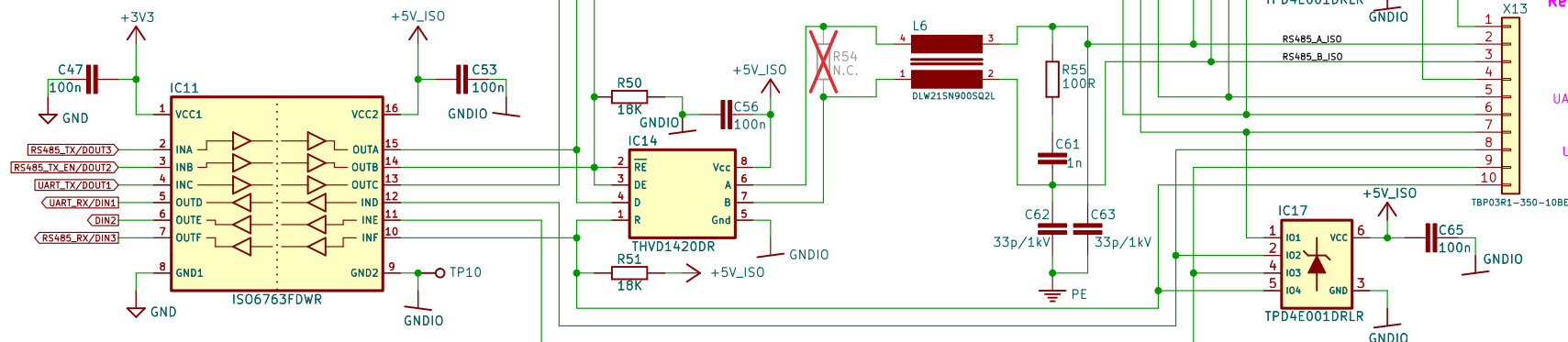
Size: A4 Date: 2026-07-25

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Rev: r1B1

Id: 4/10

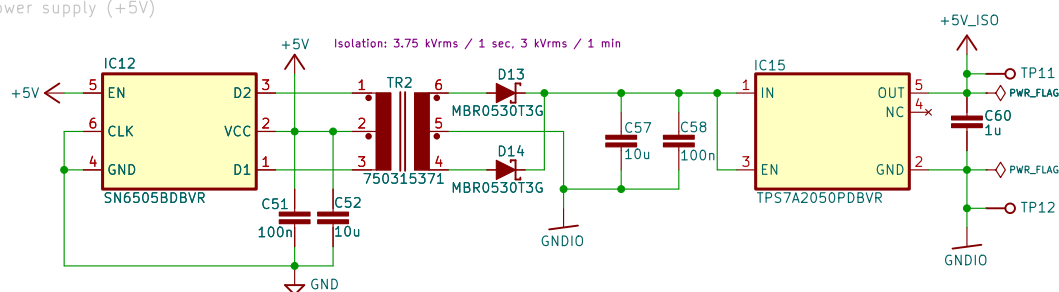
Isolated digital I/O & RS485 transceiver (half-duplex)



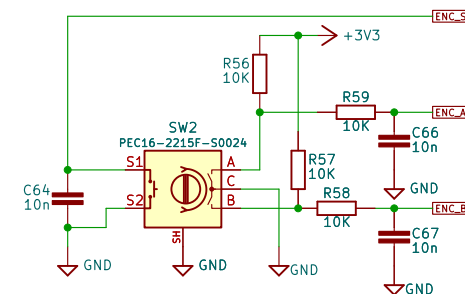
Rear panel I/O connector

- +5V_ISO 1
- RS485_A_ISO 2
- RS485_A_ISO 3
- GND_ISO 4
- UART_TX/DOUT1 5
- DOUT2 6
- DOUT3 7
- UART_RX/DIN1 8
- DIN2 9
- DIN3 10

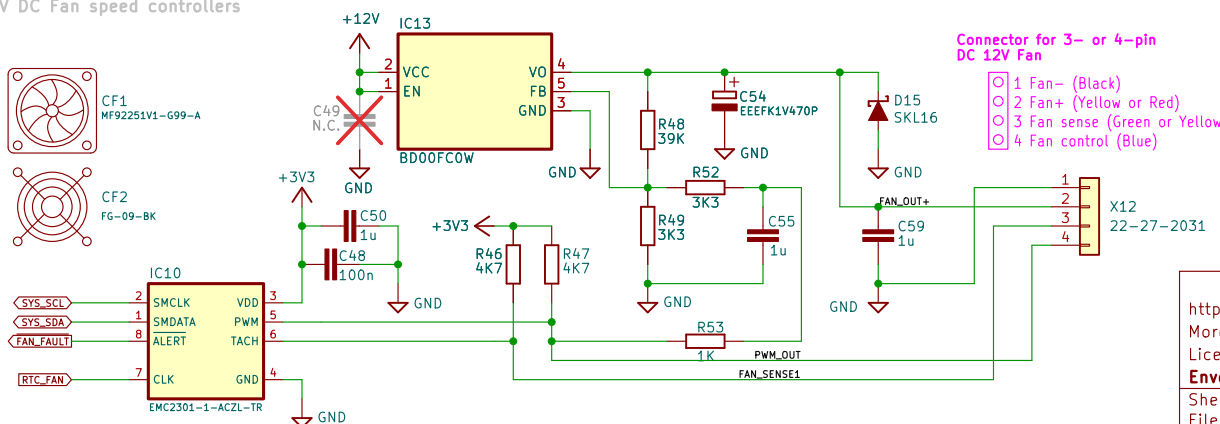
Isolated regulated power supply (+5V)



Encoder with switch



12V DC Fan speed controllers



<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

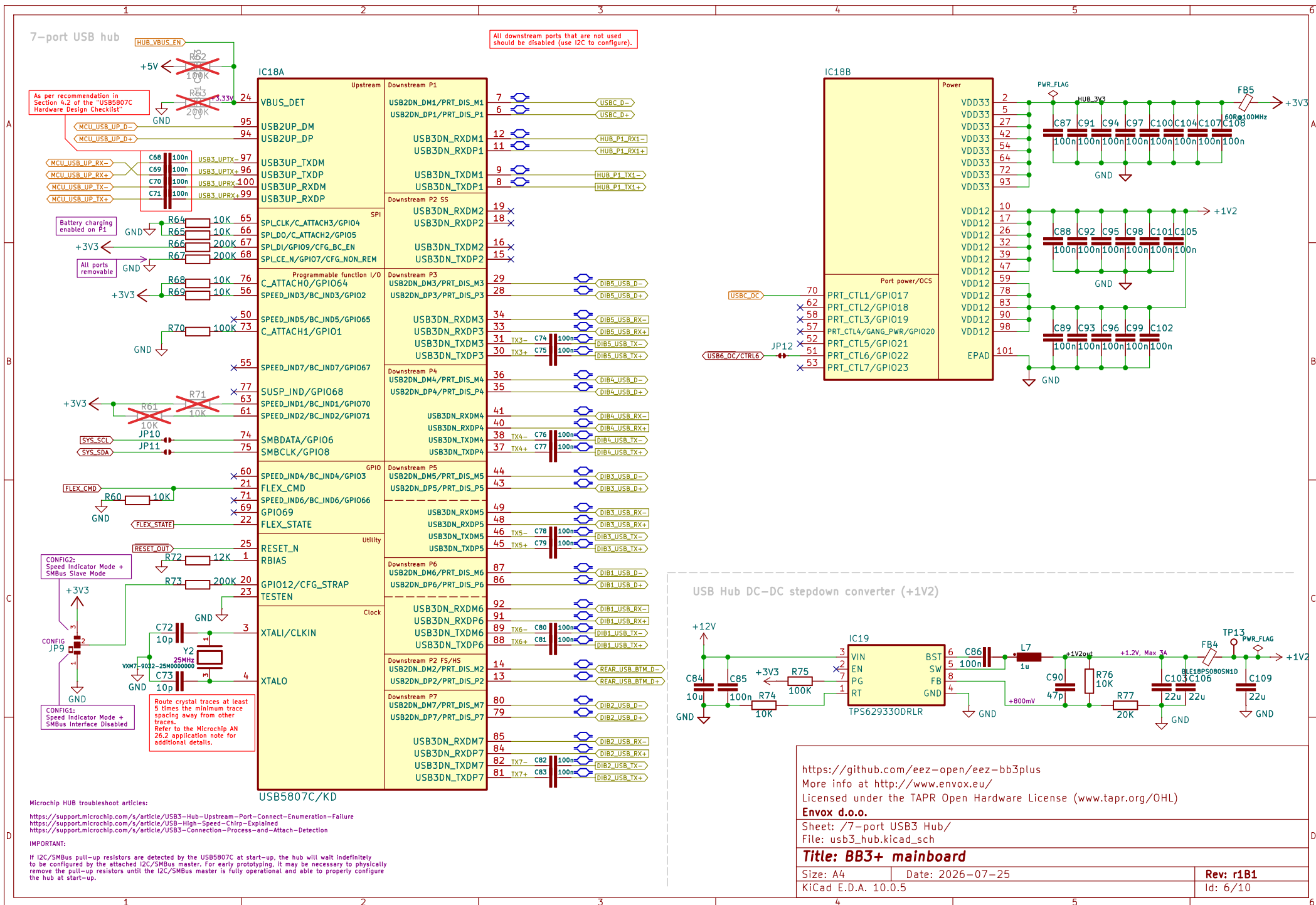
Title: BB3+ mainboard

Size: A4 Date: 2026-07-25

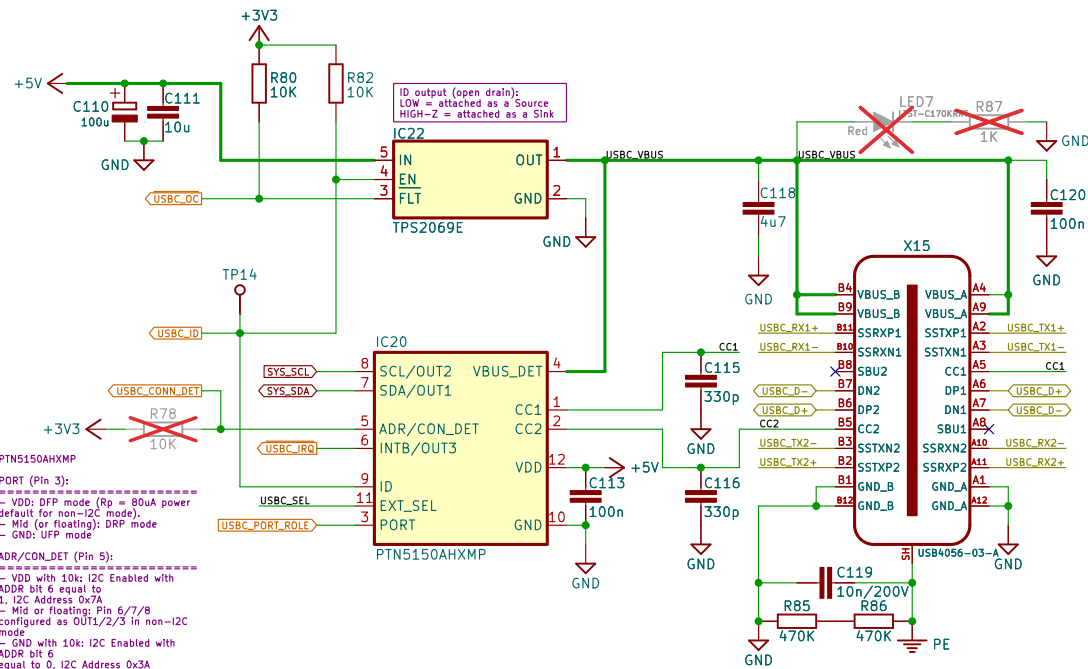
KiCad E.D.A. 10.0.5

Rev: r1B1

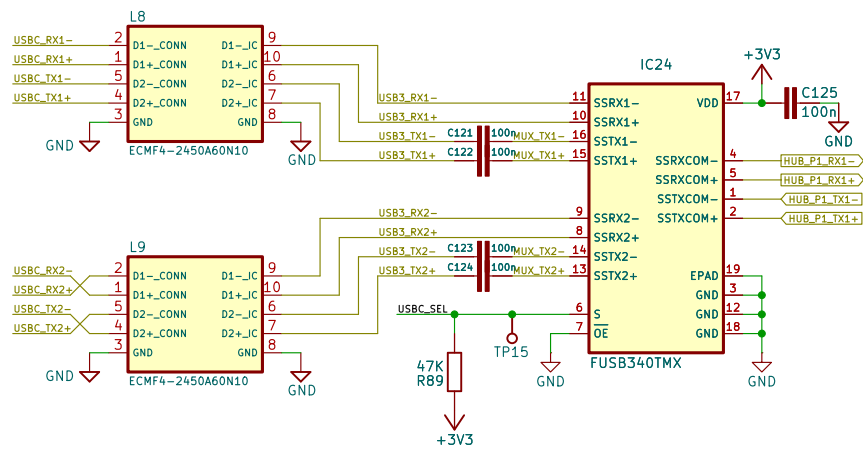
Id: 5/10



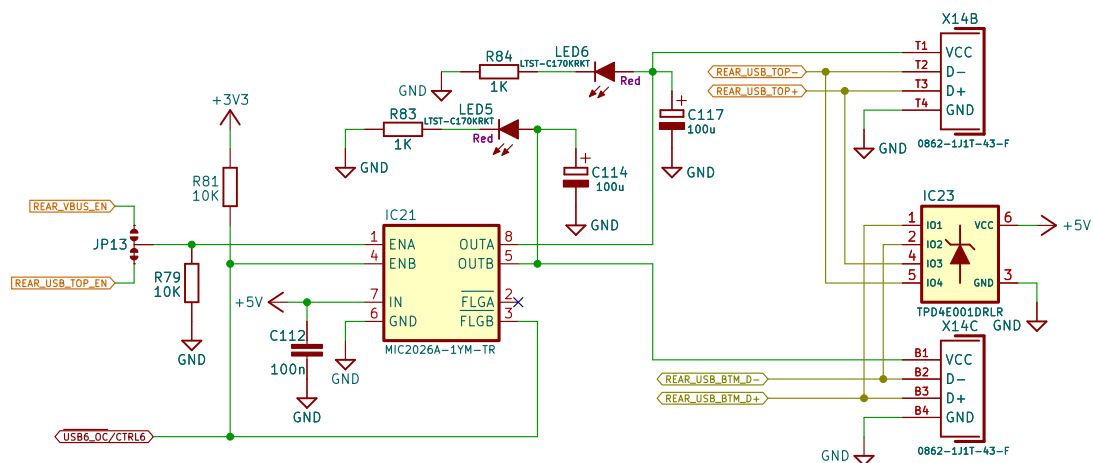
USB-C 3.0 front port (source max 1.5A)



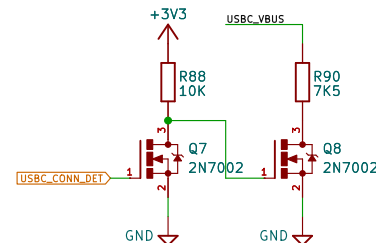
USB-C 3.0 Protection and Mux



USB2.0 Type-A (Rear panel)



VBus Discharge Circuit



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /Rear USB, USB-C/

File: usb_conn.kicad_sch

Title: BB3+ mainboard

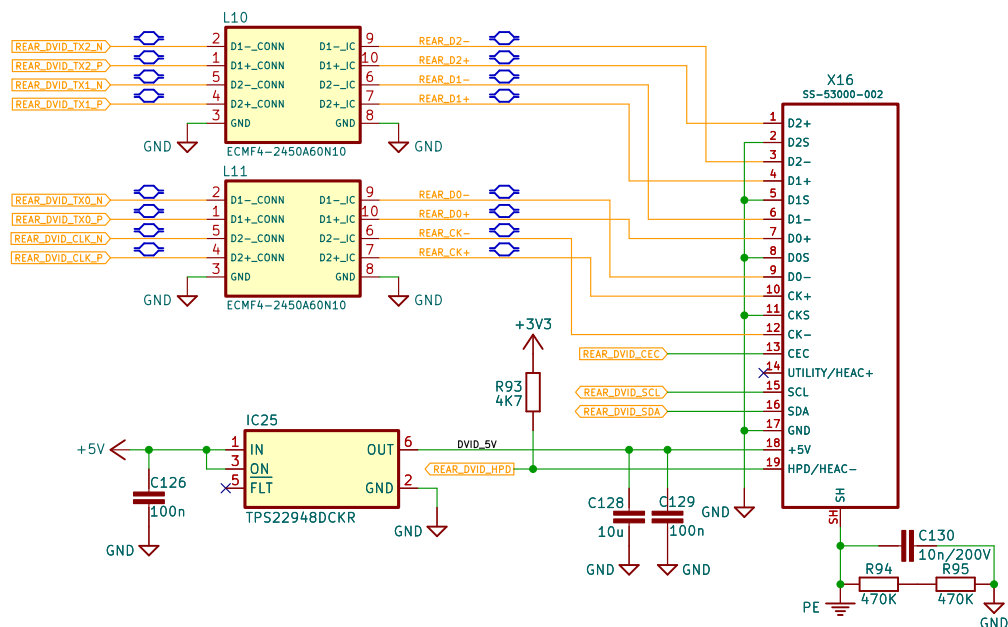
Size: A4 Date: 2026-07-25

KiCad E.D.A. 10.0.5

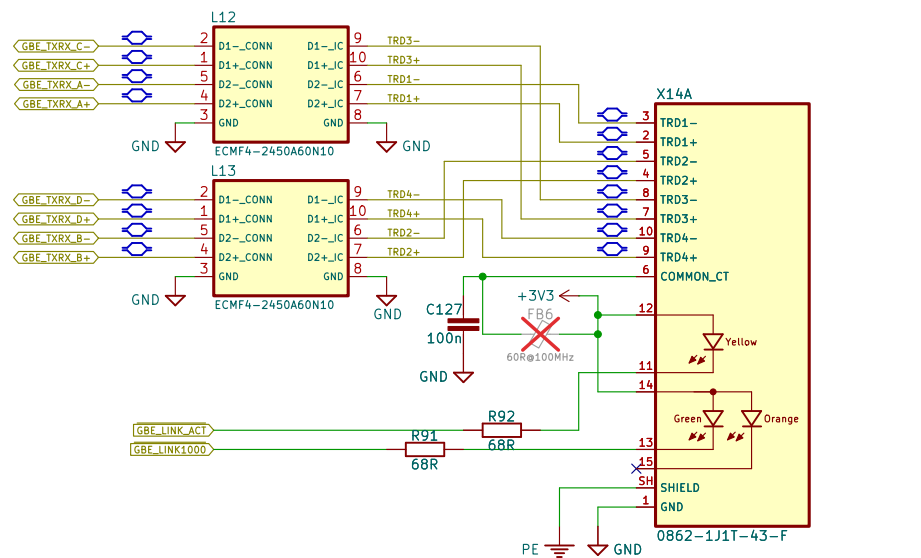
Rev: r1B1

Id: 7/10

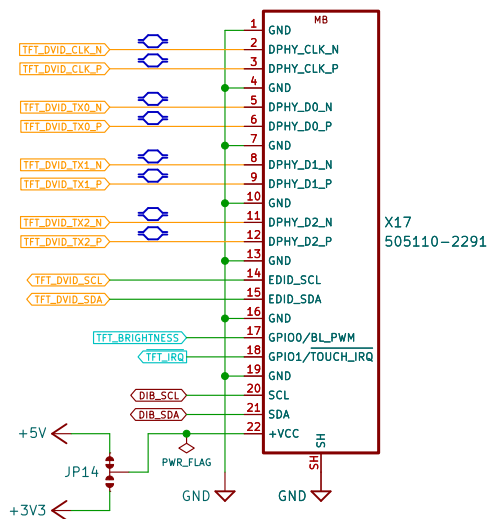
External monitor digital output



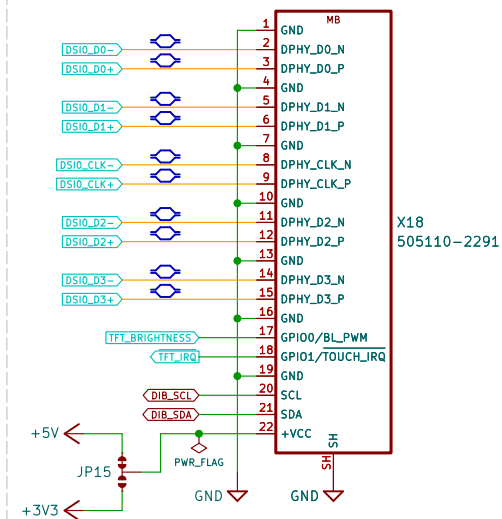
Ethernet RJ45+mag



TFT Touchscreen display DVID output (22-pin FFC)



TFT Touchscreen display DSI output (22-pin FFC)



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /Ethernet, Video outputs/

File: video_outputs.kicad_sch

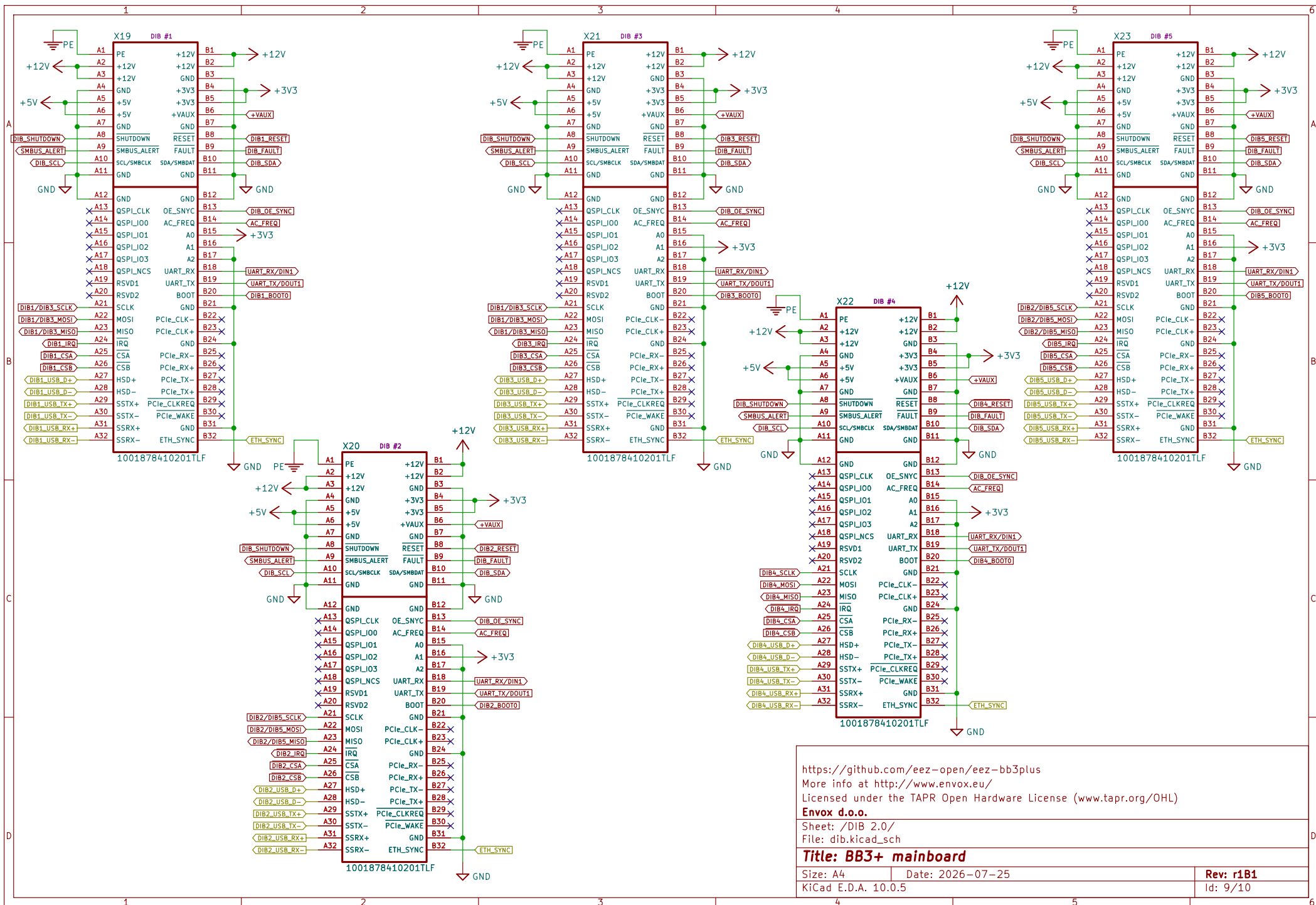
Title: BB3+ mainboard

Size: A4 Date: 2026-07-25

KiCad E.D.A. 10.0.5

Rev: r1B1

Id: 8/10



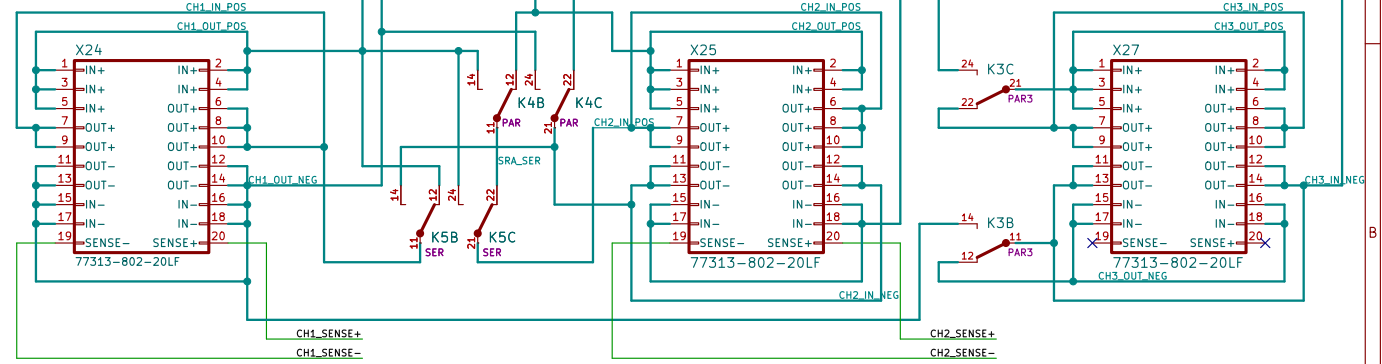
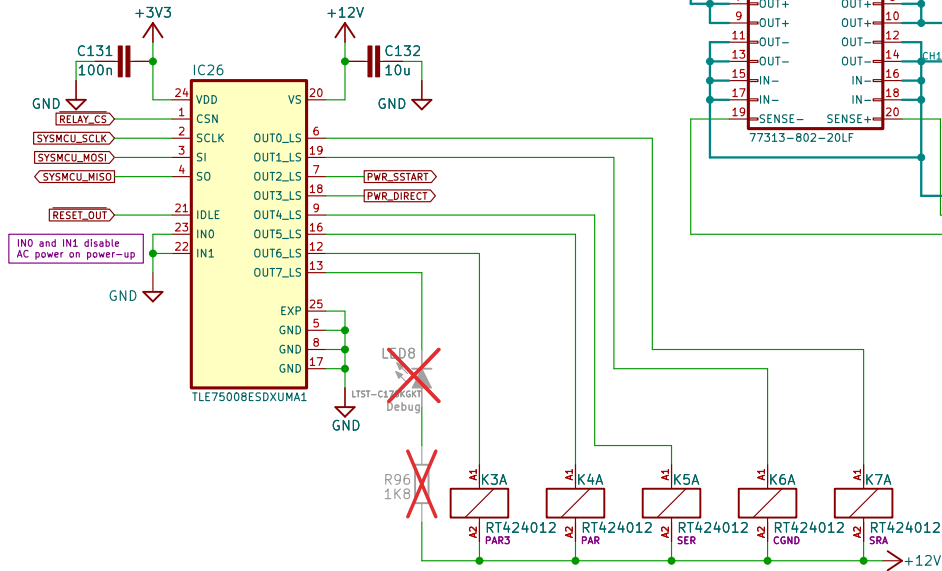
<https://github.com/eez-open/eez-bb3plus>
More info at <http://www.envox.eu/>
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Envox d.o.o.
Sheet: /DIB 2.0/
File: dib.kicad_sch

Title: BB3+ mainboard	
Size: A4	Date: 2026-07-25
KiCad E.D.A. 10.0.5	Rev: r1B1 Id: 9/10

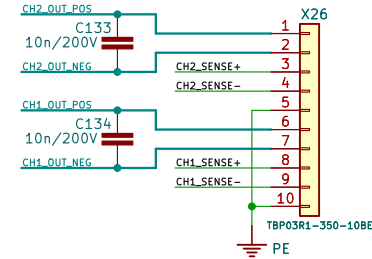
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	2 PWR+ IN
PWR+ IN 3	4 PWR+ IN
PWR+ IN 5	6 PWR+ OUT
PWR+ OUT 7	8 PWR+ OUT
PWR+ OUT 9	10 PWR+ IN
PWR- OUT 11	12 PWR- OUT
PWR- OUT 13	14 PWR- OUT
PWR- IN 15	16 PWR- IN
PWR- IN 17	18 PWR- IN
SENSE- 19	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+ 1	2
CH2_OUT- 3	4
CH2_SENSE+ 5	6
CH2_SENSE- 7	8
PE 9	10
CH1_OUT+ 11	12
CH1_OUT- 13	14
CH1_SENSE+ 15	16
CH1_SENSE- 17	18
PE 19	20

<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: /Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-07-25

KiCad E.D.A. 10.0.5

Rev: r1B1

Id: 10/10